$e^{L|t|}$. Moreover, $Y_t : \mathbb{R}^d \to \mathbb{R}^d$ is invertible and its inverse is also Lipschitz with the same constant.

Sketch of proof - We do not prove existence, which can be classically obtained by fixed-point arguments in the space of curves. As for uniqueness, and for the dependence on the initial datum x, we consider two curves $y^{(1)}$ and $y^{(2)}$, solutions of $y'(t) = \mathbf{v}_t(y(t))$, and we define $E(t) = |y^{(1)}(t) - y^{(2)}(t)|^2$. We have

$$E'(t) = 2(y^{(1)}(t) - y^{(2)}(t)) \cdot (\mathbf{v}_t(y^{(1)}(t)) - \mathbf{v}_t(y^{(2)}(t))),$$

which proves $|E'(t)| \leq 2LE(t)$. By Gronwall's Lemma, this gives $E(0)e^{-2L|t|} \leq E(t) \leq E(0)e^{2L|t|}$ and provides at the same time uniqueness, injectivity, and the bi-Lipschitz behavior of Y_t (which is also a homeomorphism from $\mathbb{R}^d$ onto $\mathbb{R}^d$ since for every $x_0 \in \mathbb{R}^d$ we can solve the ODE imposing the Cauchy datum $y(t) = x_0$).

Then, we can prove the following, that we state for simplicity in the case of Lipschitz and bounded vector fields.

Theorem 4.4. *Suppose that $\Omega \subset \mathbb{R}^d$ is either a bounded domain or $\mathbb{R}^d$ itself. Suppose that $\mathbf{v}_t : \Omega \to \mathbb{R}^d$ is Lipschitz continuous in x, uniformly in t, and uniformly bounded, and consider its flow Y_t. Suppose that, for every $x \in \Omega$ and every $t \in [0,T]$, we have $Y_t(x) \in \Omega$ (which is obvious for $\Omega = \mathbb{R}^d$, and requires suitable Neumann conditions on $\mathbf{v}_t$ otherwise). Then, for every probability $\rho_0 \in \mathscr{P}(\Omega)$ the measures $\rho_t := (Y_t)_\# \rho_0$ solve the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$ with initial datum ρ_0. Moreover, every solution of the same equation with $\rho_t \ll \mathscr{L}^d$ for every t is necessarily obtained as $\rho_t = (Y_t)_\# \rho_0$. In particular, the continuity equation admits a unique solution.*

Proof. First we check the validity of the equation when ρ_t is obtained from such a flow through (4.1). We will prove that we have a weak solution. Fix a test function $\phi \in C^1(\mathbb{R}^d)$ such that both ϕ and $\nabla\phi$ are bounded, and compute

$$\begin{aligned}\frac{d}{dt}\int_\Omega \phi \, \mathrm{d}\rho_t = \frac{d}{dt}\int \phi(y_x(t))\, \mathrm{d}\rho_0(x) &= \int_\Omega \nabla\phi(y_x(t)) \cdot y_x'(t)\, \mathrm{d}\rho_0(x) \\ &= \int_\Omega \nabla\phi(y_x(t)) \cdot \mathbf{v}_t(y_x(t))\, \mathrm{d}\rho_0(x) = \int_\Omega \nabla\phi(y) \cdot \mathbf{v}_t(y)\, \mathrm{d}\rho_t(y).\end{aligned}$$

This characterizes weak solutions.

In order to prove the second part of the statement, we first observe that (4.2), i.e.

$$\int_0^T \int_\Omega (\partial_t \phi + \mathbf{v}_t \cdot \nabla\phi)\, \mathrm{d}\rho_t\, \mathrm{d}t = 0,$$

is also valid for all Lipschitz compactly supported test functions ϕ, whenever $\rho_t \ll \mathscr{L}^d$ for every t. Indeed, if we fix a Lipschitz test function ϕ and we smooth it by convolution with a compactly supported kernel, we have a sequence $\phi_\varepsilon \in C_c^\infty$ such that $\nabla_{t,x}\phi_\varepsilon \to \nabla_{t,x}\phi$ a.e. and we can apply dominated convergence since $|\nabla_{t,x}\phi_\varepsilon| \leq \mathrm{Lip}(\phi)$ (we need ρ_t to be absolutely continuous because we only have Lebesgue-a.e. convergence).

Then we take a test function $\psi \in C^1_c(\mathbb{R}^d)$ and we define $\phi(t,x) := \psi((Y_t)^{-1}(x))$. If we can prove that $t \mapsto \int \phi(t,x)\,\mathrm{d}\rho_t(x)$ is constant, then we have proven $\rho_t = (Y_t)_\# \rho_0$. The function ϕ is Lipschitz continuous, because the flow Y_t is bi-Lipschitz; yet, it is not compactly supported in time (it is compactly supported in space since $\operatorname{spt}\phi$ is compact and, if we set $M := \sup_{t,x}|\mathbf{v}_t(x)|$, we see that $\phi(t,x)$ vanishes on all points x which are at distance larger than tM from $\operatorname{spt}\psi$). Thus, we multiply it times a cut-off function $\chi(t)$, with $\chi \in C^1_c(]0,T[)$. We have

$$\partial_t(\chi\phi) + \mathbf{v}_t \cdot \nabla(\chi\phi) = \chi'(t)\phi(t,x) + \chi(t)\left(\partial_t\phi(t,x) + \mathbf{v}_t(x)\cdot\nabla\phi(t,x)\right).$$

We can prove that, by definition of ϕ, the term $\partial_t\phi(t,x) + \mathbf{v}_t(x)\cdot\nabla\phi(t,x)$ vanishes a.e. (this corresponds to saying that ϕ is a solution of the transport equation, see Box 6.3). This is true since we have $\phi(t,T_t(x)) = \psi(x)$ and, differentiating it w.r.t. t (which is possible for a.e. (t,x)), we get $\partial_t(t,Y_t(x)) + \mathbf{v}_t(Y_t(x))\cdot\nabla\phi(t,Y_t(x)) = 0$, which means that $\partial_t\phi + \mathbf{v}_t\cdot\nabla\phi$ vanishes everywhere, as Y_t is surjective.

Hence, from the definition of distributional solution we have

$$\int_0^T \chi'(t)\mathrm{d}t \int_{\mathbb{R}^d} \phi(t,x)\,\mathrm{d}\rho_t(x) = \int_0^T \mathrm{d}t \int_{\mathbb{R}^d} \left(\partial_t(\chi\phi) + \mathbf{v}_t\cdot\nabla(\chi\phi)\right)\mathrm{d}\rho_t(x) = 0.$$

The test function $\chi \in C^1_c(]0,T[)$ being arbitrary, we get that $t \mapsto \int_{\mathbb{R}^d}\phi(t,x)\,\mathrm{d}\rho_t(x)$ is constant. □ □

4.2 Beckmann's problem

In this section we discuss a flow-minimization problem introduced by Beckmann in the '50s as a particular case of a wider class of convex optimization problem, of the form $\min\{\int H(\mathbf{w})\mathrm{d}x : \nabla\cdot\mathbf{w} = \mu - \nu\}$, for convex H. In this section we discuss the case $H(z) = |z|$, which is a limit case in the class of costs studied by Beckmann, who was indeed more interested in strictly convex costs. Strict convexity will be considered in the Discussion section 4.4.1. The reason to stick to the linear cost here, and to spend more words on it than on the general case, lies in its equivalence with the Monge Problem. We will provide all the tools to study this equivalence and this problem.

4.2.1 Introduction, formal equivalences and variants

We consider here a simple version of a problem that has been proposed by Beckmann as a model for optimal transport in the '50s. Beckmann called it *continuous transportation model* and he was not aware of Kantorovich's works and the possible links between the two theories.

Beckmann's minimal flow problem

Problem 4.5. Consider the minimization problem

$$\text{(BP)} \quad \min\left\{\int |\mathbf{w}(x)|\,\mathrm{d}x \;:\; \mathbf{w}:\Omega\to\mathbb{R}^d,\, \nabla\cdot\mathbf{w} = \mu - \nu\right\}, \tag{4.4}$$

where the divergence condition is to be read in the weak sense, with Neumann boundary conditions, i.e. $-\int \nabla\phi \cdot \mathrm{d}\mathbf{w} = \int \phi \,\mathrm{d}(\mu-\nu)$ for any $\phi \in C^1(\overline{\Omega})$.

We will see now that an equivalence between (BP) and (KP) (for the cost $c(x,y) = |x-y|$) holds true. To do that, we can look at the following considerations and formal computations.

We take the problem (BP) and re-write the constraint on $\mathbf{w}$ using

$$\sup_{\phi} \int_\Omega \nabla\phi \cdot \mathbf{w}\,\mathrm{d}x + \int_\Omega \phi \,\mathrm{d}(\mu-\nu) = \begin{cases} 0 & \text{if } \nabla\cdot\mathbf{w} = \mu-\nu, \\ +\infty & \text{otherwise.} \end{cases}$$

Hence one can write (BP) as

$$\inf_{\mathbf{w}} \left(\int_\Omega |\mathbf{w}|\,\mathrm{d}x + \sup_{\phi} \int_\Omega \nabla\phi\cdot\mathbf{w}\,\mathrm{d}x + \int_\Omega \phi\,\mathrm{d}(\mu-\nu) \right)$$
$$= \sup_{\phi} \left(\int_\Omega \phi\,\mathrm{d}(\mu-\nu) + \inf_{\mathbf{w}} \int_\Omega (|\mathbf{w}| - \nabla\phi\cdot\mathbf{w})\,\mathrm{d}x \right), \quad (4.5)$$

where inf and sup have been exchanged formally as in the computations of Chapter 1. Then, one notes that

$$\inf_{\mathbf{w}} \int_\Omega (|\mathbf{w}| - \nabla\phi\cdot\mathbf{w})\,\mathrm{d}x = \begin{cases} 0 & \text{if } |\nabla\phi| \le 1 \\ -\infty & \text{otherwise} \end{cases}$$

and this leads to the dual formulation for (BP) which gives

$$\sup \left\{ \int_\Omega \phi\,\mathrm{d}(\mu-\nu) \quad : \quad |\nabla\phi| \le 1 \right\}.$$

Since this problem is exactly the same as (DP) (a consequence of the fact that Lip_1 functions are exactly those functions whose gradient is smaller than 1), this provides a formal equivalence between (BP) and (KP). We say that it is only formal because we did not prove the equality in (4.5). Note that we also need to suppose that Ω is convex, otherwise functions with gradient smaller than 1 are only Lip_1 according to the geodesic distance in Ω.

Most of the considerations above, especially those on the problem (BP), are specific to the cost equal to the distance $|x-y|$. In general, for costs of the form $h(x-y)$, h needs to be 1-homogeneous, if one wants some similar result to hold. We refer to [196, 197] for a general way to transform convex costs into convex 1-homogeneous ones (by adding one variable, corresponding to time)[1]. An interesting generalization which keeps the same formalism of the case of the Euclidean distance concerns a cost c which comes from a Riemannian distance $k(x)$.

[1] In this way the flow-minimization problem corresponding to costs of the form $|x-y|^p$ is transformed into the so-called Benamou-Brenier problem, that we will discuss in Chapters 5 and 6, but we do not push this analogy to further conclusions.

Consider indeed

$$\min\left\{\int k(x)|\mathbf{w}(x)|\,\mathrm{d}x\ :\ \nabla\cdot\mathbf{w}=\mu-\nu\right\}$$

which corresponds, by duality with the functions u such that $|\nabla u|\leq k$, to

$$\min\left\{\int d_k(x,y)\,\mathrm{d}\gamma(x,y)\ :\ \gamma\in\Pi(\mu,\nu)\right\},$$

where $d_k(x,y)=\inf_{\omega(0)=x,\,\omega(1)=y}L_k(\omega):=\int_0^1 k(\omega(t))|\omega'(t)|\mathrm{d}t$ is the distance associated to the Riemannian metric k.

The generalization above is suitable when we want to model a non-uniform cost for the movement (due to geographical obstacles or configurations). Such a model is not always satisfying, for instance in urban transport, where we want to consider the fact that the metric k is usually not a priori known, but it depends on the traffic distribution itself. We will develop this aspect in the discussion section 4.4.1, together with a completely different problem which is somehow "opposite" (Section 4.4.2): instead of looking at transport problems where concentration of the mass is penalized (because of traffic congestion), we look at problems where it is indeed encouraged, because of the so-called "economy of scale" (i.e. the larger the mass you transport, the cheaper the individual cost).

4.2.2 Producing a minimizer for the Beckmann Problem

The first remark on Problem (BP) is that a priori it is not well-posed, in the sense that there could not exist an L^1 vector field minimizing the L^1 norm under divergence constraints. This is easy to understand if we think at the direct method in Calculus of Variations to prove existence: we take a minimizing sequence $\mathbf{w}_n$ and we would like to extract a converging subsequence. Could we do this, from $\mathbf{w}_n\rightharpoonup\mathbf{w}$ it would be easy to prove that $\mathbf{w}$ still satisfies $\nabla\cdot\mathbf{w}=\mu-\nu$, since the relation

$$-\int\nabla\phi\cdot\mathbf{w}_n\,\mathrm{d}x=\int\phi\,\mathrm{d}(\mu-\nu)$$

would pass to the limit as $n\to\infty$. Yet, the information $\int|\mathbf{w}(x)|\,\mathrm{d}x\leq C$ is not enough to extract a converging sequence, even weakly. Indeed, the space L^1 being non-reflexive, bounded sequences are not guaranteed to have weakly converging subsequences. This is on the contrary the case for dual spaces (and for reflexive spaces, which are roughly speaking the dual of their dual).

Note that the strictly convex version that was proposed by Beckmann and that we will review in Section 4.4.1 is much better to handle: if for instance we minimize $\int|\mathbf{w}|^2\,\mathrm{d}x$, then we can use weak compactness in L^2, which is way easier to have than compactness in L^1.

To avoid this difficulty, we will choose the natural setting for (BP), i.e. the framework of vector measures.

Box 4.2. **– Memo** – Vector measures

Definition - A finite vector measure λ on a set Ω is a map associating to every Borel subset $A \subset \Omega$ a value $\lambda(A) \in \mathbb{R}^d$ such that, for every disjoint union $A = \bigcup_i A_i$ (with $A_i \cap A_j = \emptyset$ for $i \neq j$), we have

$$\sum_i |\lambda(A_i)| < +\infty \quad \text{and} \quad \lambda(A) = \sum_i \lambda(A_i).$$

We denote by $\mathscr{M}^d(\Omega)$ the set of finite vector measures on Ω. To such measures we can associate a positive scalar measure $|\lambda| \in \mathscr{M}_+(\Omega)$ through

$$|\lambda|(A) := \sup\left\{\sum_i |\lambda(A_i)| \ : \ A = \bigcup_i A_i \ \text{ with } A_i \cap A_j = \emptyset \text{ for } i \neq j\right\}.$$

This scalar measure is called *total variation measure* of λ. Note that for simplicity we only consider the Euclidean norm on $\mathbb{R}^d$, and write $|\lambda|$ instead of $||\lambda||$ (a notation that we keep for the total mass of the total variation measure, see below), but the same could be defined for other norms as well.

The integral of a Borel function $\xi : \Omega \to \mathbb{R}^d$ w.r.t. λ is well-defined if $|\xi| \in L^1(\Omega, |\lambda|)$, is denoted $\int \xi \cdot \mathrm{d}\lambda$ and can be computed as $\sum_{i=1}^d \int \xi_i \, \mathrm{d}\lambda_i$, thus reducing to integrals of scalar functions according to scalar measures. It could also be defined as a limit of integral of piecewise constant functions.

Functional analysis facts - The quantity $||\lambda|| := |\lambda|(\Omega)$ is a norm on $\mathscr{M}^d(\Omega)$, and this normed space is the dual of $C_0(\Omega;\mathbb{R}^d)$, the space of continuous function on Ω vanishing at infinity, through the duality $(\xi,\lambda) \mapsto \int \xi \cdot \mathrm{d}\lambda$. This gives a notion of $\overset{*}{\rightharpoonup}$ convergence for which bounded sets in $\mathscr{M}^d(\Omega)$ are compact. As for scalar measures, we denote by $\rightharpoonup$ the weak convergence in duality with C_b functions.

A clarifying fact is the following.

Proposition - For every $\lambda \in \mathscr{M}^d(\Omega)$ there exists a Borel function $u : \Omega \to \mathbb{R}^d$ such that $\lambda = u \cdot |\lambda|$ and $|u| = 1$ a.e. (for the measure $|\lambda|$). In particular, $\int \xi \cdot \mathrm{d}\lambda = \int (\xi \cdot u) \, \mathrm{d}|\lambda|$.

Sketch of proof - The existence of a function u is a consequence, via Radon-Nikodym Theorem, of $\lambda \ll |\lambda|$ (every A set such that $|\lambda|(A) = 0$ obviously satisfies $\lambda(A) = 0$). The condition $|u| = 1$ may be proven by considering the sets $\{|u| < 1 - \varepsilon\}$ and $\{u \cdot e > a + \varepsilon\}$ for all hyperplane such that the unit ball B_1 is contained in $\{x \in \mathbb{R}^d \ : \ x \cdot e \leq a\}$ (and, actually, we have $B_1 = \bigcap_{e,a}\{x \in \mathbb{R}^d \ : \ x \cdot e \leq a\}$, the intersection being reduced to a countable intersection). These sets must be negligible otherwise we have a contradiction on the definition of $|\lambda|$.

With these definitions in mind, we can prove the following theorem. We denote by $\mathscr{M}^d_{\mathrm{div}}$ the space of vector measures with divergence which is a scalar measure.

Theorem 4.6. *Suppose that Ω is a compact convex domain in $\mathbb{R}^d$. Then, the problem*

$$\text{(BP)} \quad \min\left\{|\mathbf{w}|(\Omega) \ : \ \mathbf{w} \in \mathscr{M}^d_{\mathrm{div}}(\Omega)\,,\, \nabla \cdot \mathbf{w} = \mu - \nu\right\}$$

(with divergence imposed in the weak sense, i.e. for every $\phi \in C^1(\overline{\Omega})$ we impose $-\int \nabla\phi \cdot \mathrm{d}\mathbf{w} = \int \phi \, \mathrm{d}(\mu - \nu)$, which also includes homogeneous Neumann boundary

conditions) admits a solution. Moreover, its minimal value equals the minimal value of (KP) *and a solution of* (BP) *can be built from a solution of* (KP). *The two problems are hence equivalent.*

Proof. The first point that we want to prove is the equality of the minimal values of (BP) and (KP). We start from min(BP)$\geq$ min(KP). In order to do so, take an arbitrary function $\phi \in C^1$ with $|\nabla\phi| \leq 1$. Consider that for any $\mathbf{w}$ with $\nabla \cdot \mathbf{w} = \mu - \nu$, we have

$$|\mathbf{w}|(\Omega) = \int_\Omega 1\,\mathrm{d}|\mathbf{w}| \geq \int_\Omega (-\nabla\phi)\cdot \mathrm{d}\mathbf{w} = \int_\Omega \phi\,\mathrm{d}(\mu-\nu).$$

If one takes a sequence of $\mathrm{Lip}_1 \cap C^1$ functions uniformly converging to the Kantorovich potential u such that $\int_\Omega u\,\mathrm{d}(\mu-\nu) = \max(\mathrm{DP}) = \min(\mathrm{KP})$ (for instance take convolutions $\phi_k = \eta_k * u$) then we get

$$\int_\Omega \mathrm{d}|\mathbf{w}| \geq \min(\mathrm{KP})$$

for any admissible $\mathbf{w}$, i.e. min(BP)$\geq$ min(KP).

We will show at the same time the reverse inequality and how to construct an optimal $\mathbf{w}$ from an optimal γ for (KP).

Actually, one way to produce a solution to this divergence-constrained problem, is the following (see [71]): take an optimal transport plan γ and build a vector measure $\mathbf{w}_{[\gamma]}$ defined[2] through

$$\langle \mathbf{w}_{[\gamma]}, \xi\rangle := \int_{\Omega\times\Omega}\int_0^1 \omega'_{x,y}(t)\cdot\xi(\omega_{x,y}(t))\mathrm{d}t\;\mathrm{d}\gamma(x,y),$$

for every $\xi \in C^0(\Omega;\mathbb{R}^d)$, $\omega_{x,y}$ being a parametrization of the segment $[x,y]$ (it is clear that this is the point where convexity of Ω is needed). Even if for this proof it would not be important, we will fix the constant speed parametrization, i.e. $\omega_{x,y}(t) = (1-t)x + ty$.

It is not difficult to check that this measure satisfies the divergence constraint, since if one takes $\xi = \nabla\phi$ then

$$\int_0^1 \omega'_{x,y}(t)\cdot\xi(\omega_{x,y}(t))\,\mathrm{d}t = \int_0^1 \frac{d}{dt}\left(\phi(\omega_{x,y}(t)\right)\mathrm{d}t = \phi(y) - \phi(x)$$

and hence $\langle \mathbf{w}_{[\gamma]}, \nabla\phi\rangle = \int \phi\, d(\nu-\mu)$ and $\mathbf{w}_{[\gamma]} \in \mathscr{M}^d_{\mathrm{div}}(\Omega)$.

To estimate its mass we can see that $|\mathbf{w}_{[\gamma]}| \leq \sigma_\gamma$, where the scalar measure σ_γ is defined through

$$\langle \sigma_\gamma, \phi\rangle := \int_{\Omega\times\Omega}\int_0^1 |\omega'_{x,y}(t)|\phi(\omega_{x,y}(t))\mathrm{d}t\;\mathrm{d}\gamma, \qquad \text{for all } \phi \in C^0(\Omega;\mathbb{R})$$

[2] The strange notation $\mathbf{w}_{[\gamma]}$ is chosen so as to distinguish from the object $\mathbf{w}_Q$ that we will introduce in Section 4.2.3.

and it is called *transport density*. Actually, we can even say more, since we can use the Kantorovich potential u (see Chapter 3), and write[3]

$$\omega'_{x,y}(t) = -|x-y|\frac{x-y}{|x-y|} = -|x-y|\nabla u(\omega_{x,y}(t)).$$

This is valid for every $t \in]0,1[$ and every $x,y \in \mathrm{spt}(\gamma)$ (so that $\omega_{x,y}(t)$ is in the interior of the transport ray $[x,y]$, if $x \neq y$; anyway for $x = y$, both expression vanish).

This allow to write, for every $\xi \in C^0(\Omega;\mathbb{R}^d)$

$$\begin{aligned}\langle \mathbf{w}_{[\gamma]}, \xi \rangle &= \int_{\Omega\times\Omega}\int_0^1 -|x-y|\nabla u(\omega_{x,y}(t))\cdot\xi(\omega_{x,y}(t))\mathrm{d}t\,\mathrm{d}\gamma(x,y)\\ &= -\int_0^1 \mathrm{d}t\int_{\Omega\times\Omega}\nabla u(\omega_{x,y}(t))\cdot\xi(\omega_{x,y}(t))|x-y|\,\mathrm{d}\gamma(x,y)\end{aligned}$$

If we introduce the function $\pi_t : \Omega\times\Omega \to \Omega$ given by $\pi_t(x,y) = \omega_{x,y}(t) = (1-t)x+ty$, we get

$$\langle \mathbf{w}_{[\gamma]}, \xi \rangle = -\int_0^1 \mathrm{d}t\int_\Omega \nabla u(z)\cdot\xi(z)\,\mathrm{d}\big((\pi_t)_\#(c\cdot\gamma)\big),$$

where $c\cdot\gamma$ is the measure on $\Omega\times\Omega$ with density $c(x,y) = |x-y|$ w.r.t. γ.

Since on the other hand the same kind of computations give

$$\langle \sigma_\gamma, \phi \rangle = \int_0^1 \mathrm{d}t\int_\Omega \phi(z)\,\mathrm{d}\big((\pi_t)_\#(c\cdot\gamma)\big), \tag{4.6}$$

we get $\langle \mathbf{w}_{[\gamma]}, \xi \rangle = \langle \sigma_\gamma, -\xi\cdot\nabla u\rangle$, which shows

$$\mathbf{w}_{[\gamma]} = -\nabla u\cdot\sigma_\gamma.$$

This gives the density of $\mathbf{w}_{[\gamma]}$ with respect to σ_γ and confirms $|\mathbf{w}_{[\gamma]}| \leq \sigma_\gamma$.

The mass of σ_γ is obviously

$$\int_\Omega \mathrm{d}\sigma_\gamma = \int_\Omega\int_0^1 |\omega'_{x,y}(t)|\mathrm{d}t\,\mathrm{d}\gamma(x,y) = \int_{\Omega\times\Omega}|x-y|\,\mathrm{d}\gamma(x,y) = \min(\mathrm{KP}),$$

which proves the optimality of $\mathbf{w}_{[\gamma]}$ since no other $\mathbf{w}$ may do better than this, thanks to the first part of the proof. This also proves $\min(\mathrm{BP}) = \min(\mathrm{KP})$. □ □

Monge-Kantorovich system and transport density The scalar measure σ_γ that we have just defined is called *transport density*. It has been been introduced in [160] and [72, 71] for different goals. In [72, 71] it is connected to some shape-optimization problems that we will not develop in this book. On the other hand, in [160] it has been used to provide one of the first solutions to the Monge problem with cost $|x-y|$.

[3] Pay attention to the use of the gradient of the Kantorovich potential u: we are using the result of Lemma 3.6 which provides differentiability of u on transport rays.

The important fact is that the measure $\sigma = \sigma_\gamma$ solves, together with the Kantorovich potential u, the so-called Monge-Kantorovich system

$$\begin{cases} \nabla\cdot(\sigma\nabla u) = \mu - \nu & \text{in } \Omega \\ |\nabla u| \leq 1 & \text{in } \Omega, \\ |\nabla u| = 1 & \sigma - \text{a.e.} \end{cases} \tag{4.7}$$

This system is a priori only solved in a formal way, since the product of the L^∞ function ∇u with the measure σ has no meaning if $\sigma \notin L^1$. To overcome this difficulty there are two possibilities: either we pass through the theory of σ-tangential gradient (see for instance [146] or [72]), or we give conditions to have $\sigma_\gamma \in L^1$. This second choice is what we do in Section 4.3, where we also show better L^p estimates.

As we said, the solution of (4.7) has been used in [160] to find a transport map T, optimal for the cost $|x-y|$. This map was defined as the flow of a certain vector field depending on σ and ∇u, exactly in the spirit of the Dacorogna-Moser construction that we will see in Section 4.2.3. However, because of the lack of regularity of σ and ∇u, the Beckmann problem $\min\left\{\int|\mathbf{w}| \,:\, \nabla\cdot\mathbf{w} = \mu-\nu\right\}$, whose solution is $\mathbf{w} = \sigma_\gamma\nabla u$, was approximated through $\min\left\{\int|\mathbf{w}|^p \,:\, \nabla\cdot\mathbf{w} = \mu-\nu\right\}$, for $p>1$. We will see in Section 4.4.1 that this problem, connected to traffic congestion issues, is solved by a vector field obtained from the solution of a $\Delta_{p'}$ equation, and admits some regularity results. In this way, the dual of the Monge problem is approximated by p'-Laplacian equations, for $p' \to \infty$.

4.2.3 Traffic intensity and traffic flows for measures on curves

We introduce in this section some objects that generalize both $\mathbf{w}_{[\gamma]}$ and σ_γ and that will be useful for many goals. They will be used both for proving the characterization of the optimal $\mathbf{w}$ as coming from an optimal plan γ and for the modeling issues of the discussion section. As a byproduct, we also present a new proof of a decomposition result by Smirnov, [287] (see also [278]).

Let us introduce some notations. We define the set AC(Ω) as the set of absolutely continuous curves $\omega : [0,1] \mapsto \Omega$. We suppose Ω to be compact, with non-empty interior, in the whole section. Given $\omega \in \mathrm{AC}(\Omega)$ and a continuous function ϕ, let us set

$$L_\phi(\omega) := \int_0^1 \phi(\omega(t))|\omega'(t)|\mathrm{d}t. \tag{4.8}$$

This quantity is the length of the curve, weighted with the weight ϕ. When we take $\phi = 1$ we get the usual length of ω and we denote it by $L(\omega)$ instead of $L_1(\omega)$. Note that these quantities are well-defined since $\omega \in \mathrm{AC}(\Omega)$ implies that ω is differentiable a.e. and $\omega' \in L^1([0,1])$ (we recall the definition of AC curves: they are curves ω with their distributional derivative $\omega' \in L^1$ and $\omega(t_1) - \omega(t_0) = \int_{t_0}^{t_1}\omega'(t)\mathrm{d}t$ for every $t_0 < t_1$). For simplicity, from now on we will write $\mathscr{C}$ (the space of "curves") for AC(Ω), when there is no ambiguity on the domain.

We consider probability measures Q on the space $\mathscr{C}$. We restrict ourselves to measures Q such that $\int L(\omega)\,\mathrm{d}Q(\omega) < +\infty$: these measures will be called *traffic plans*, according to a terminology introduced in [50]. We endow the space $\mathscr{C}$ with the uniform convergence. Note that Ascoli-Arzelà's theorem guarantees that the sets $\{\omega \in \mathscr{C} : \mathrm{Lip}(\omega) \leq \ell\}$ are compact (for the uniform convergence) for every ℓ. We will associate two measures on Ω to such a Q. The first is a scalar one, called *traffic intensity* and denoted by $i_Q \in \mathscr{M}_+(\Omega)$; it is defined by

$$\int_\Omega \phi \, \mathrm{d}i_Q := \int_{\mathscr{C}} \Big(\int_0^1 \phi(\omega(t))|\omega'(t)|\mathrm{d}t \Big) \mathrm{d}Q(\omega) = \int_{\mathscr{C}} L_\phi(\omega)\mathrm{d}Q(\omega).$$

for all $\phi \in C(\Omega, \mathbb{R}_+)$. This definition (taken from [115]) is a generalization of the notion of transport density. The interpretation is the following: for a subregion A, $i_Q(A)$ represents the total cumulated traffic in A induced by Q, i.e. for every path we compute "how long" does it stay in A, and then we average on paths.

We also associate a vector measure $\mathbf{w}_Q$ to any traffic plan $Q \in \mathscr{P}(\mathscr{C})$ via

$$\forall \xi \in C(\Omega; \mathbb{R}^d) \quad \int_\Omega \xi \cdot \mathrm{d}\mathbf{w}_Q := \int_{\mathscr{C}} \left(\int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t \right) \mathrm{d}Q(\omega).$$

We will call $\mathbf{w}_Q$ *traffic flow* induced by Q. Taking a gradient field $\xi = \nabla\phi$ in the previous definition yields

$$\int_\Omega \nabla\phi \cdot \mathrm{d}\mathbf{w}_Q = \int_{\mathscr{C}} [\phi(\omega(1)) - \phi(\omega(0))]\mathrm{d}Q(\omega) = \int_\Omega \phi \, \mathrm{d}((e_1)_\# Q - (e_0)_\# Q)$$

(we recall that e_t denotes the evaluation map at time t, i.e. $e_t(\omega) := \omega(t)$).

From now on, we will restrict our attention to *admissible* traffic plans Q, i.e. traffic plans such that $(e_0)_\# Q = \mu$ and $(e_1)_\# Q = \nu$, where μ and ν are two prescribed probability measures on Ω. This means that

$$\nabla \cdot \mathbf{w}_Q = \mu - \nu$$

and hence $\mathbf{w}_Q$ is an admissible flow connecting μ and ν. Note that the divergence is always considered in a weak (distributional) sense, and automatically endowed with Neumann boundary conditions, i.e. when we say $\nabla \cdot \mathbf{w} = f$ we mean $\int \nabla\phi \cdot \mathrm{d}\mathbf{w} = -\int \phi \, \mathrm{d}f$ for all $\phi \in C^1(\Omega)$, without any condition on the boundary behavior of the test function ϕ.

Coming back to $\mathbf{w}_Q$, it is easy to check that $|\mathbf{w}_Q| \leq i_Q$, where $|\mathbf{w}_Q|$ is the total variation measure of the vector measure $\mathbf{w}_Q$. This last inequality is in general not an equality, since the curves of Q could produce some cancellations (imagine a non-negligible amount of curves passing through the same point with opposite directions, so that $\mathbf{w}_Q = 0$ and $i_Q > 0$).

We need some properties of the traffic intensity and traffic flow.

Proposition 4.7. *Both $\mathbf{w}_Q$ and i_Q are invariant under reparametrization (i.e. if $\mathrm{T} : \mathscr{C} \to \mathscr{C}$ is a map such that for every ω the curve $\mathrm{T}(\omega)$ is just a reparametrization in time of ω, then $\mathbf{w}_{\mathrm{T}_\# Q} = \mathbf{w}_Q$ and $i_{\mathrm{T}_\# Q} = i_Q$).*

For every Q, the total mass $i_Q(\Omega)$ equals the average length of the curves according to Q, i.e. $\int_{\mathscr{C}} L(\omega)\,\mathrm{d}Q(\omega) = i_Q(\Omega)$. In particular, $\mathbf{w}_Q$ and i_Q are finite measures thanks to the definition of traffic plan.

If $Q_n \rightharpoonup Q$ and $i_{Q_n} \rightharpoonup i$, then $i \geq i_Q$.

If $Q_n \rightharpoonup Q$, $\mathbf{w}_{Q_n} \rightharpoonup \mathbf{w}$ and $i_{Q_n} \rightharpoonup i$, then $||\mathbf{w}-\mathbf{w}_Q|| \leq i(\Omega) - i_Q(\Omega)$. In particular, if $Q_n \rightharpoonup Q$ and $i_{Q_n} \rightharpoonup i_Q$, then $\mathbf{w}_{Q_n} \rightharpoonup \mathbf{w}_Q$.

Proof. The invariance by reparametrization comes from the invariance of both $L_\phi(\omega)$ and $\int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t$.

The formula $\int_{\mathscr{C}} L(\omega)\,\mathrm{d}Q(\omega) = i_Q(\Omega)$ is obtained from the definition of i_Q by testing against the function 1.

To check the inequality $i \geq i_Q$, fix a positive test function $\phi \in C(\Omega)$ and suppose $\phi \geq \varepsilon_0 > 0$. Write

$$\int_\Omega \phi\,\mathrm{d}i_{Q_n} = \int_{\mathscr{C}} \left(\int_0^1 \phi(\omega(t))|\omega'(t)|\mathrm{d}t \right) \mathrm{d}Q_n(\omega). \tag{4.9}$$

Note that the function $\mathscr{C} \ni \omega \mapsto L_\phi(\omega) = \int_0^1 \phi(\omega(t))|\omega'(t)|\mathrm{d}t$ is positive and lower-semi-continuous w.r.t. ω. Indeed, if we take a sequence $\omega_n \to \omega$, from the bound $\phi \geq \varepsilon_0 > 0$ we can assume that $\int |\omega_n'(t)|\mathrm{d}t$ is bounded. Then we can infer $\omega_n' \rightharpoonup \omega'$ weakly (as measures, or in L^1), which implies, up to subsequences, the existence of a measure $f \in \mathscr{M}_+([0,1])$ such that $f \geq |\omega'|$ and $|\omega_n'| \rightharpoonup f$. Moreover, $\phi(\omega_n(t)) \to \phi(\omega(t))$ uniformly, which gives $\int \phi(\omega_n(t))|\omega_n'(t)|\mathrm{d}t \to \int \phi(\omega(t))\mathrm{d}f(t) \geq \int \phi(\omega(t))|\omega'(t)|\mathrm{d}t$.

This allows to pass to the limit in (4.9), thus obtaining

$$\int \phi\,\mathrm{d}i = \lim_n \int \phi\,\mathrm{d}i_{Q_n} = \liminf_n \int_{\mathscr{C}} L_\phi(\omega)\mathrm{d}Q_n(\omega) \geq \int_{\mathscr{C}} L_\phi(\omega)\mathrm{d}Q(\omega) = \int \phi\,\mathrm{d}i_Q.$$

If we take an arbitrary test function ϕ (without a lower bound), add a constant ε_0 and apply the previous reasoning we get $\int(\phi+\varepsilon_0)\mathrm{d}i \geq \int(\phi+\varepsilon_0)\mathrm{d}i_Q$. Letting $\varepsilon_0 \to 0$, as i is a finite measure and ϕ is arbitrary, we get $i \geq i_Q$.

To check the last property, fix a smooth vector test function ξ and a number $\lambda > 1$ and look at

$$\begin{aligned}\int_\Omega \xi \cdot \mathrm{d}\mathbf{w}_{Q_n} &= \int_{\mathscr{C}} \left(\int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t \right) \mathrm{d}Q_n(\omega) \\ &= \int_{\mathscr{C}} \left(\int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t + \lambda|\xi||_\infty L(\omega) \right) \mathrm{d}Q_n(\omega) - \lambda||\xi||_\infty i_{Q_n}(\Omega), \end{aligned} \tag{4.10}$$

where we just added and subtracted the total mass of i_{Q_n}, equal to the average of $L(\omega)$ according to Q_n. Now note that

$$\mathscr{C} \ni \omega \mapsto \int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t + \lambda||\xi||_\infty L(\omega) \geq (\lambda-1)||\xi||_\infty L(\omega)$$

is l.s.c. in ω (use the same argument as above, noting that, if we take $\omega_n \to \omega$, we may assume $L(\omega_n)$ to be bounded, and obtain $\omega_n' \rightharpoonup \omega'$). This means that if we pass to the limit in (4.10) we get

$$\begin{aligned}\int_\Omega \xi \cdot \mathrm{d}\mathbf{w} &= \lim_n \int_\Omega \xi \cdot \mathrm{d}\mathbf{w}_{Q_n} \\ &\geq \int_{\mathscr{C}} \left(\int_0^1 \xi(\omega(t)) \cdot \omega'(t)\mathrm{d}t + \lambda ||\xi||_\infty L(\omega) \right) \mathrm{d}Q(\omega) - \lambda ||\xi||_\infty i(\Omega) \\ &= \int_\Omega \xi \cdot \mathrm{d}\mathbf{w}_Q + \lambda ||\xi||_\infty (i_Q(\Omega) - i(\Omega)).\end{aligned}$$

By replacing ξ with $-\xi$ we get

$$\left| \int_\Omega \xi \cdot \mathrm{d}\mathbf{w} - \int_\Omega \xi \cdot \mathrm{d}\mathbf{w}_Q \right| \leq \lambda ||\xi||_\infty (i(\Omega) - i_Q(\Omega)).$$

Letting $\lambda \to 1$ and taking the sup over ξ with $||\xi||_\infty \leq 1$ we get the desired estimate $||\mathbf{w} - \mathbf{w}_Q|| \leq i(\Omega) - i_Q(\Omega)$.

The very last property is evident: indeed one can assume up to a subsequence that $\mathbf{w}_{Q_n} \rightharpoonup \mathbf{w}$ holds for a certain $\mathbf{w}$, and $i = i_Q$ implies $\mathbf{w} = \mathbf{w}_Q$ (which also implies the full convergence of the sequence). □ □

Box 4.3. **– Good to know!** – Dacorogna-Moser transport

Let us present here a particular case of the ideas from [133] (first used in optimal transport in [160]):

Construction - Suppose that $\mathbf{w} : \Omega \to \mathbb{R}^d$ is a Lipschitz vector field with $\mathbf{w} \cdot \mathbf{n} = 0$ on $\partial\Omega$ and $\nabla \cdot \mathbf{w} = f_0 - f_1$, where f_0, f_1 are positive probability densities which are Lipschitz continuous and bounded from below. Then we can define the non-autonomous vector field $\mathbf{v}_t(x)$ via

$$\mathbf{v}_t(x) = \frac{\mathbf{w}(x)}{f_t(x)} \quad \text{where } f_t = (1-t)f_0 + tf_1$$

and consider the Cauchy problem

$$\begin{cases} y_x'(t) = \mathbf{v}_t(y_x(t)) \\ y_x(0) = x \end{cases},$$

We define a map $Y : \Omega \to \mathscr{C}$ by associating to every x the curve $Y(x)$ given by $y_x(\cdot)$. Then, we look for the measure $Q = Y_\# f_0$ and $\rho_t := (e_t)_\# Q := (Y_t)_\# f_0$. Thanks to the consideration in Section 4.1.2, ρ_t solves the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$. Yet, it is easy to check that f_t also solves the same equation since $\partial_t f_t = f_1 - f_0$ and $\nabla \cdot (\mathbf{v}_t f_t) = \nabla \cdot \mathbf{w} = f_0 - f_1$. By the uniqueness result of Section 4.1.2, from $\rho_0 = f_0$ we infer $\rho_t = f_t$. In particular, $x \mapsto y_x(1)$ is a transport map from f_0 to f_1.

It it interesting to compute the traffic intensity and the traffic flow associated to the measure Q in Dacorogna-Moser construction. Fix a scalar test function ϕ:

$$\int_\Omega \phi\, \mathrm{d}i_Q = \int_\Omega \int_0^1 \phi(y_x(t))|\mathbf{v}_t(y_x(t))|\mathrm{d}t f_0(x)\,\mathrm{d}x$$
$$= \int_0^1 \int_\Omega \varphi(y)|\mathbf{v}_t(y)|f_t(y)\mathrm{d}y\mathrm{d}t = \int_\Omega \phi(y)|\mathbf{w}(y)|\mathrm{d}y$$

so that $i_Q = |\mathbf{w}|$. Analogously, fix a vector test function ξ

$$\int_\Omega \xi \cdot d\mathbf{w}_Q = \int_\Omega \int_0^1 \xi(y_x(t)) \cdot \mathbf{v}_t(y_x(t))\mathrm{d}t f_0(x)\,\mathrm{d}x$$
$$= \int_0^1 \int_\Omega \xi(y) \cdot \mathbf{v}_t(y) f_t(y)\mathrm{d}y\mathrm{d}t = \int_\Omega \xi(y) \cdot \mathbf{w}(y)\mathrm{d}y,$$

which shows $\mathbf{w}_Q = \mathbf{w}$. Note that in this case we have $|\mathbf{w}_Q| = i_Q$ and this is due to the fact that no cancellation is possible, since all the curves share the same direction at every given point.

With these tools we want to provide a proof of a decomposition result by Smirnov ([287]), for which we need an extra approximation result.

Lemma 4.8. *Consider two probabilities $\mu, \nu \in \mathscr{P}(\Omega)$ on a smooth compact domain Ω and a vector measure $\mathbf{w} \in \mathscr{M}^d_{\mathrm{div}}$ satisfying $\nabla \cdot \mathbf{w} = \mu - \nu$ in distributional sense (with homogeneous Neumann boundary conditions). Then, for every smooth compact domain Ω' containing Ω in its interior, there exist a family of vector fields $\mathbf{w}^\varepsilon \in C^\infty(\Omega')$ with $\mathbf{w}^\varepsilon \cdot \mathbf{n}_{\Omega'} = 0$, and two families of densities $\mu^\varepsilon, \nu^\varepsilon \in C^\infty(\Omega')$, bounded from below by positive constants $k_\varepsilon > 0$, with $\nabla \cdot \mathbf{w}^\varepsilon = \mu^\varepsilon - \nu^\varepsilon$ and $\int_{\Omega'} \mu^\varepsilon = \int_{\Omega'} \nu^\varepsilon = 1$, weakly converging to $\mathbf{w}, \mu$ and ν as measures, respectively, and satisfying $|\mathbf{w}^\varepsilon| \rightharpoonup |\mathbf{w}|$.*

Proof. First, extend μ, ν and $\mathbf{w}$ at 0 out of Ω and take convolutions (in the whole space $\mathbb{R}^d$) with a gaussian kernel η_ε, so that we get $\hat{\mathbf{w}}^\varepsilon := \mathbf{w} * \eta_\varepsilon$ and $\hat{\mu}^\varepsilon := \mu * \eta_\varepsilon$, $\hat{\nu}^\varepsilon := \nu * \eta_\varepsilon$, still satisfying $\nabla \cdot \hat{\mathbf{w}}^\varepsilon = \mu^\varepsilon - \nu^\varepsilon$. Since the Gaussian kernel is strictly positive, we also have strictly positive densities for $\hat{\mu}^\varepsilon$ and $\hat{\nu}^\varepsilon$. These convolved densities and vector field would do the job required by the theorem, but we have to take care of the support (which is not Ω') and of the boundary behavior.

Let us set $\int_{\Omega'} \hat{\mu}^\varepsilon = 1 - a_\varepsilon$ and $\int_{\Omega'} \hat{\nu}^\varepsilon = 1 - b_\varepsilon$. It is clear that $a_\varepsilon, b_\varepsilon \to 0$ as $\varepsilon \to 0$. Consider also $\hat{\mathbf{w}}^\varepsilon \cdot \mathbf{n}_{\Omega'}$: due to $d(\Omega, \partial\Omega') > 0$ and to the fact that η_ε goes uniformly to 0 locally outside the origin, we also have $|\hat{\mathbf{w}}^\varepsilon \cdot \mathbf{n}_{\Omega'}| \le c_\varepsilon$, with $c_\varepsilon \to 0$.

Consider u^ε the solution to

$$\begin{cases} \Delta u^\varepsilon = \frac{a_\varepsilon - b_\varepsilon}{|\Omega'|} & \text{inside } \Omega' \\ \frac{\partial u^\varepsilon}{\partial \mathbf{n}} = -\hat{\mathbf{w}}^\varepsilon \cdot \mathbf{n} & \text{on } \partial\Omega', \\ \int_{\Omega'} u^\varepsilon = 0 & \end{cases}$$

and the vector field $\delta^\varepsilon = \nabla u^\varepsilon$. Note that a solution exists thanks to $-\int_{\partial\Omega'} \hat{\mathbf{w}}^\varepsilon \cdot \mathbf{n}_{\Omega'} = a_\varepsilon - b_\varepsilon$. Note also that an integration by parts shows

$$\int_{\Omega'} |\nabla u^\varepsilon|^2 = -\int_{\partial\Omega'} u^\varepsilon (\hat{\mathbf{w}}^\varepsilon \cdot \mathbf{n}_{\Omega'}) - \int_{\Omega'} u^\varepsilon \left(\frac{a_\varepsilon - b_\varepsilon}{|\Omega'|} \right) \le C ||\nabla u^\varepsilon||_{L^2}(c_\varepsilon + a_\varepsilon + b_\varepsilon),$$

and provides $||\nabla u^\varepsilon||_{L^2} \le C(a_\varepsilon + b_\varepsilon + c_\varepsilon) \to 0$. This shows $||\delta^\varepsilon||_{L^2} \to 0$.

Now take

$$\mu^\varepsilon = \hat{\mu}^\varepsilon \llcorner \Omega' + \frac{a_\varepsilon}{|\Omega'|}; \quad \nu^\varepsilon = \hat{\nu}^\varepsilon \llcorner \Omega' + \frac{b_\varepsilon}{|\Omega'|}; \quad \mathbf{w}^\varepsilon = \hat{\mathbf{w}}^\varepsilon \llcorner \Omega' + \delta^\varepsilon,$$

and check that all the requirements are satisfied. In particular, the last one is satisfied since $||\delta^\varepsilon||_{L^1} \to 0$ and $|\hat{\mathbf{w}}^\varepsilon| \rightharpoonup |\mathbf{w}|$ by general properties of the convolutions. □ □

Remark 4.9. Note that considering explicitly the dependence on Ω' it is also possible to obtain the same statement with a sequence of domains Ω'_ε converging to Ω (for instance in the Hausdorff topology). It is just necessary to choose them so that, setting $t_\varepsilon := d(\Omega, \partial\Omega'_\varepsilon)$, we have $||\eta_\varepsilon||_{L^\infty(B(0,t_\varepsilon)^c)} \to 0$. For the Gaussian kernel, this is satisfied whenever $t_\varepsilon^2/\varepsilon \to \infty$ and can be guaranteed by taking $t_\varepsilon = \varepsilon^{1/3}$.

With these tools we can now prove

Theorem 4.10. *For every finite vector measure* $\mathbf{w} \in \mathscr{M}^d_{\mathrm{div}}(\Omega)$ *and* $\mu, \nu \in \mathscr{P}(\Omega)$ *with* $\nabla \cdot \mathbf{w} = \mu - \nu$ *there exists a traffic plan* $Q \in \mathscr{P}(\mathscr{C})$ *with* $(e_0)_\# Q = \mu$ *and* $(e_1)_\# Q = \nu$ *such that* $|\mathbf{w}_Q| = i_Q \le |\mathbf{w}|$, *and* $||\mathbf{w} - \mathbf{w}_Q|| + ||\mathbf{w}_Q|| = ||\mathbf{w} - \mathbf{w}_Q|| + i_Q(\Omega) = ||\mathbf{w}||$. *In particular we have* $|\mathbf{w}_Q| \ne |\mathbf{w}|$ *unless* $\mathbf{w}_Q = \mathbf{w}$.

Proof. By means of Lemma 4.8 and Remark 4.9 we can produce an approximating sequence $(\mathbf{w}^\varepsilon, \mu^\varepsilon, \nu^\varepsilon) \rightharpoonup (\mathbf{w}, \mu, \nu)$ of C^∞ functions supported on domains Ω_ε converging to Ω. We apply Dacorogna-Moser's construction to this sequence of vector fields, thus obtaining a sequence of measures Q_ε. We can consider these measures as probability measures on $\mathscr{C} := \mathrm{AC}(\Omega')$ (where $\Omega \subset \Omega_\varepsilon \subset \Omega'$) which are, each, concentrated on curves valued in Ω_ε. They satisfy $i_{Q_\varepsilon} = |\mathbf{w}^\varepsilon|$ and $\mathbf{w}_{Q_\varepsilon} = \mathbf{w}^\varepsilon$. We can reparametrize by constant speed the curves on which Q_ε is supported, without changing traffic intensities and traffic flows. This means that we use curves ω such that $L(\omega) = \mathrm{Lip}(\omega)$. The equalities

$$\int_{\mathscr{C}} \mathrm{Lip}(\omega)\, \mathrm{d}Q_\varepsilon(\omega) = \int_{\mathscr{C}} L(\omega)\, \mathrm{d}Q_\varepsilon(\omega) = \int_{\Omega'} i_{Q_\varepsilon} = \int_{\Omega'} |\mathbf{w}^\varepsilon| \to |\mathbf{w}|(\Omega') = |\mathbf{w}|(\Omega)$$

show that $\int_{\mathscr{C}} \mathrm{Lip}(\omega)\, \mathrm{d}Q_\varepsilon(\omega)$ is bounded and hence Q_ε is tight (since the sets $\{\omega \in \mathscr{C} : \mathrm{Lip}(\omega) \le L\}$ are compact). Hence, up to subsequences, we can assume $Q_\varepsilon \rightharpoonup Q$. The measure Q is obviously concentrated on curves valued in Ω. The measures Q_ε were constructed so that $(e_0)_\# Q_\varepsilon = \mu^\varepsilon$ and $(e_1)_\# Q_\varepsilon = \nu^\varepsilon$, which implies, at the limit, $(e_0)_\# Q = \mu$ and $(e_1)_\# Q = \nu$. Moreover, thanks to Proposition 4.7, since $i_{Q_\varepsilon} = |\mathbf{w}_\varepsilon| \rightharpoonup |\mathbf{v}|$ and $\mathbf{w}_{Q_\varepsilon} \rightharpoonup \mathbf{w}$, we get $|\mathbf{w}| \ge i_Q \ge |\mathbf{w}_Q|$ and $||\mathbf{w} - \mathbf{w}_Q|| \le |\mathbf{w}|(\Omega) - i_Q(\Omega)$. This gives $||\mathbf{w} - \mathbf{w}_Q|| + ||\mathbf{w}_Q|| \le ||\mathbf{w} - \mathbf{w}_Q|| + i_Q(\Omega) \le ||\mathbf{w}||$ and the opposite inequality $||\mathbf{w}|| \le ||\mathbf{w} - \mathbf{w}_Q|| + ||\mathbf{w}_Q||$ is always satisfied. □ □

Remark 4.11. The previous statement contains a milder version of Theorem C in [287], i.e. the decomposition of any $\mathbf{w}$ into a cycle $\mathbf{w} - \mathbf{w}_Q$ (we call *cycle* all divergence-free vector measures) and a flow $\mathbf{w}_Q$ induced by a measure on paths, with $||\mathbf{w}|| = ||\mathbf{w} - \mathbf{w}_Q|| + ||\mathbf{w}_Q||$. The only difference with the theorem in [287] is the fact that it guarantees that one can choose Q concentrated on *simple* curves, which we did not take care of here (on the other hand, **Ex**(27) provides a partial solution to this issue).

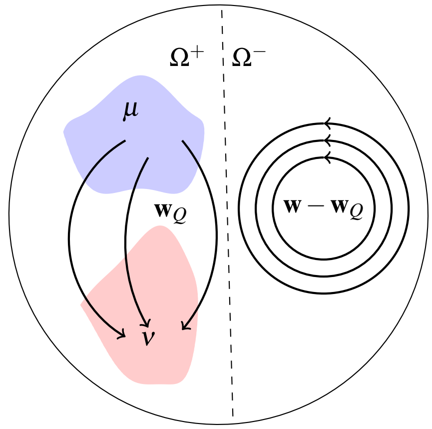

Figure 4.1: The decomposition of $\mathbf{w}$ in the example of Remark 4.12.

Remark 4.12. It is possible to guess what happens to a cycle through this construction. Imagine the following example (as in Figure 4.2.3): Ω is composed of two parts Ω^+ and Ω^-, with $\mathrm{spt}(\mu)\cup\mathrm{spt}(\nu)\subset\Omega^+$ and a cycle of $\mathbf{w}$ is contained in Ω^-. When we build the approximations μ^ε and ν^ε, we will have positive mass in Ω^-, but very small. Because of the denominator in the definition of $\mathbf{v}$, the curves produced by Dacorogna-Moser will follow this cycle very fast, passing many times on each point of the cycle. Hence, the flow $\mathbf{w}^\varepsilon$ in Ω^- is obtained from a very small mass which passes many times. This implies that the measure Q_ε of this set of curves will disappear at the limit $\varepsilon\to 0$. Hence, the measure Q will be concentrated on curves staying in Ω^+ and $\mathbf{w}_Q=0$ on Ω^-. However, in this way we got rid of that particular cycle, but the same does not happen for cycles located in regions with positive masses of μ and ν. In particular nothing guarantees that $\mathbf{w}_Q$ has no cycles.

4.2.4 Beckman problem in one dimension

The one-dimensional case is very easy in what concerns Beckmann formulation of the optimal transport problem, but it is interesting to analyze it: indeed, it allows to check the consistency with the Monge formulation and to use the results throughout next sections. We will take $\Omega=[a,b]\subset\mathbb{R}$.

First of all, note that the condition $\nabla\cdot\mathbf{w}=\mu-\nu$ is much stronger in dimension one than in higher dimension. Indeed, the divergence is the trace of the Jacobian matrix, and hence prescribing it only gives one constraint on a matrix which has a priori $d\times d$ degrees of freedom. On the contrary, in dimension one there is only one partial derivative for the vector field $\mathbf{w}$ (which is actually a scalar), and this completely prescribes the behavior of $\mathbf{w}$. Indeed, the condition $\nabla\cdot\mathbf{w}=\mu-\nu$ with Neumann boundary conditions implies that $\mathbf{w}$ must be the antiderivative of $\mu-\nu$ with $\mathbf{w}(a)=0$ (the fact that μ and ν have the same mass also implies $\mathbf{w}(b)=0$). Note that the fact that its derivative is a measure gives $\mathbf{w} \in BV([a,b])$ (we can say that $\mathscr{M}^d_{\mathrm{div}} = BV$ when $d = 1$).

Box 4.4. **– Memo** – Bounded variation functions in one variable

BV functions are defined as L^1 functions whose distributional derivatives are measures. In dimension one this has a lot of consequences. In particular these functions coincide a.e. with functions which have bounded total variation in a pointwise sense: for each $f : [a,b] \to \mathbb{R}$ define

$$TV(f;[a,b]) := \sup\left\{\sum_{i=0}^{N-1} |f(t_{i+1}) - f(t_i)| \;:\; a = t_0 < t_1 < t_2 < \cdots < t_N = b\right\}.$$

Functions of bounded variation are defined as those f such that $TV(f;[a,b]) < \infty$. It is easy to check that BV functions form a vector space, and that monotone functions are BV (indeed, if f is monotone we have $TV(f;[a,b]) = |f(b) - f(a)|$. Lipschitz functions are also BV and $TV(f;[a,b]) \le \mathrm{Lip}(f)(b-a)$. On the other hand, continuous functions are not necessarily BV (but absolutely continuous functions are BV), neither it is the case for differentiable functions (obviously, C^1 functions, which are Lipschitz on bounded intervals, are BV). As an example one can consider

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x^2}\right) & \text{if } x \neq 0 \\ 0 & \text{for } x = 0, \end{cases}$$

which is differentiable everywhere but not BV.

On the other hand, BV functions have several properties:

Properties of BV functions in $\mathbb{R}$ - If $TV(f;[a,b]) < \infty$ then f is the difference of two monotone functions (in particular we can write $f(x) = TV(f;[a,x]) - (TV(f;[a,x]) - f(x))$, both terms being non-decreasing functions); it is a bounded function and $\sup f - \inf f \le TV(f;[a,b])$; it has the same continuity and differentiability properties of monotone functions (it admits left and right limits at every point, it is continuous up to a countable set of points and differentiable a.e.).

In particular, in 1D, we have $BV \subset L^\infty$ which is not the case in higher dimension (in general, we have $BV \subset L^{d/(d-1)}$).

We finish by stressing the connections with measures: for every positive measure μ on $[a,b]$ we can build a monotone function by taking its cumulative distribution function, i.e. $F_\mu(x) = \mu([a,x])$ and the distributional derivative of this function is exactly the measure μ. Conversely, every monotone increasing function on a compact interval is the cumulative distribution function of a (unique) positive measure, and every BV function is the cumulative distribution function of a (unique) signed measure.

As a consequence, we have the following facts:

- In dimension one, there is only one competitor $\mathbf{w}$ which is given by $\mathbf{w} = F_\mu - F_\nu$ with $F_\mu(x) = \mu([a,x])$ and $F_\nu(x) = \nu([a,x])$.
- This field $\mathbf{w}$ belongs to $BV([a,b])$ and hence to every L^p space, including L^∞.
- Possible higher regularity of $\mathbf{w}$ depends on the regularity of $\mu - \nu$ (for instance, we have $\mathbf{w} \in W^{1,p}$ whenever $\mu, \nu \in L^p$).

- The minimal cost in Beckmann's problem is given by $||F_\mu - F_\nu||_{L^1}$, which is consistent with Proposition 2.17.
- The transport density σ, characterized by $\mathbf{w} = -u' \cdot \sigma$ is given by $\sigma = |\mathbf{w}|$ and shares the same summability properties of $\mathbf{w}$; it also belongs to BV as a composition of a BV function with the absolute value function.

4.2.5 Characterization and uniqueness of the optimal w

In this section we will show two facts: first we prove that the optimal $\mathbf{w}$ in the (BP) always comes from an optimal transport plan γ and then we prove that all the optimal γ give the same $\mathbf{w}_{[\gamma]}$ and the same σ_γ, provided one of the two measures is absolutely continuous.

Theorem 4.13. *Let* $\mathbf{w}$ *be optimal in* (BP)*: then there is an optimal transport plan* γ *such that* $\mathbf{w} = \mathbf{w}_{[\gamma]}$.

Proof. Thanks to Theorem 4.10, we can find a measure $Q \in \mathscr{P}(\mathscr{C})$ with $(e_0)_\# Q = \mu$ and $(e_1)_\# Q = \nu$ such that $|\mathbf{w}| \geq |\mathbf{w}_Q|$ Yet, the optimality of $\mathbf{w}$ implies the equality $|\mathbf{w}| = |\mathbf{w}_Q|$ and the same Theorem 4.10 gives in such a case $\mathbf{w} = \mathbf{w}_Q$, as well as $|\mathbf{w}| = i_Q$. We can assume Q to be concentrated on curves parametrized by constant speed. Define $S : \Omega \times \Omega \to C$ the map associating to every pair (x,y) the segment $\omega_{x,y}$ parametrized with constant speed: $\omega_{x,y}(t) = (1-t)x + ty$. The statement is proven if we can prove that $Q = S_\# \gamma$ with γ an optimal transport plan.

Indeed, using again the optimality of $\mathbf{w}$ and Theorem 4.10, we get

$$\begin{aligned}\min(\mathrm{BP}) = |\mathbf{w}|(\Omega) = i_Q(\Omega) = \int_{\mathscr{C}} L(\omega)\,\mathrm{d}Q(\omega) \geq \int_{\mathscr{C}} |\omega(0) - \omega(1)|\,\mathrm{d}Q(\omega) \\ = \int_{\Omega\times\Omega} |x-y|\,\mathrm{d}((e_0,e_1)_\# Q)(x,y) \geq \min(\mathrm{KP}).\end{aligned}$$

The equality $\min(\mathrm{BP}) = \min(\mathrm{KP})$ implies that all these inequalities are equalities. In particular Q must be concentrated on curves such that $L(\omega) = |\omega(0) - \omega(1)|$, i.e. segments. Also, the measure $(e_0, e_1)_\# Q$, which belongs to $\Pi(\mu,\nu)$, must be optimal in (KP). This concludes the proof. □ □

The proof of the following result is essentially taken from [8].

Theorem 4.14. *If* $\mu \ll \mathscr{L}^d$*, then the vector field* $\mathbf{w}_{[\gamma]}$ *does not depend on the choice of the optimal plan* γ.

Proof. Let us fix a Kantorovich potential u for the transport between μ and ν. This potential does not depend on the choice of γ. It determines a partition into transport rays: Corollary 3.8 guarantees that the only points of Ω which belong to several transport rays are non-differentiability points for u, and are hence Lebesgue-negligible. Let us call S the set of points which belong to several transport rays: we have $\mu(S) = 0$, but we do not suppose $\nu(S) = 0$ (ν is not supposed to be absolutely continuous). However, γ is concentrated on $(\pi_x)^{-1}(S^c)$. We can then disintegrate (see Section 2.3) γ

according to the transport ray containing the point x. More precisely, we define a map $R:\Omega\times\Omega\to\mathscr{R}$, valued in the set $\mathscr{R}$ of all transport rays, sending each pair (x,y) into the ray containing x. This is well-defined γ-a.e. and we can write $\gamma=\gamma^r\otimes\lambda$, where $\lambda=R_\#\gamma$ and we denote by r the variable related to transport rays. Note that, for a.e. $r\in\mathscr{R}$, the plan γ^r is optimal between its own marginals (otherwise we could replace it with an optimal plan, do it in a measurable way, and improve the cost of γ).

The measure $\mathbf{w}_{[\gamma]}$ may also be obtained through this disintegration, and we have $\mathbf{w}_{[\gamma]}=\mathbf{w}_{\gamma^r}\otimes\lambda$. This means that, in order to prove that $\mathbf{w}_{[\gamma]}$ does not depend on γ, we just need to prove that each $\mathbf{w}_{\gamma^r}$ and the measure λ do not depend on it. For the measure λ this is easy: it has been obtained as an image measure through a map only depending on x, and hence only depends on μ. Concerning $\mathbf{w}_{\gamma^r}$, note that it is obtained in the standard Beckmann way from an optimal plan, γ^r. Hence, thanks to the considerations in Section 4.2.4 about the 1D case, it uniquely depends on the marginal measures of this plan.

This means that we only need to prove that $(\pi_x)_\#\gamma^r$ and $(\pi_y)_\#\gamma^r$ do not depend on γ. Again, this is easy for $(\pi_x)_\#\gamma^r$, since it must coincide with the disintegration of μ according to the map R (by uniqueness of the disintegration). It is more delicate for the second marginal.

The second marginal $\nu^r:=(\pi_y)_\#\gamma^r$ will be decomposed into two parts:

$$(\pi_y)_\#(\gamma^r_{|\Omega\times S})\;+\;(\pi_y)_\#(\gamma^r_{|\Omega\times S^c}).$$

This second part coincides with the disintegration of $\nu_{|S^c}$, which obviously does not depend on γ (since it only depends on the set S, which is built upon u).

We need now to prove that $\nu^r_{|S}=(\pi_y)_\#(\gamma^r_{|\Omega\times S})$ does not depend on γ. Yet, this measure can only be concentrated on the two endpoints of the transport ray r, since these are the only points where different transport rays can meet. This means that this measure is purely atomic and composed by at most two Dirac masses. Not only, the endpoint where u is maximal (i.e. the first one for the order on the ray, see Section 3.1) cannot contain some mass of ν: indeed the transport must follow a precise direction on each transport ray (as a consequence of $u(x)-u(y)=|x-y|$ on $\mathrm{spt}(\gamma)$), and the only way to have some mass of the target measure at the "beginning" of the transport ray would be to have an atom for the source measure as well. Yet, μ is absolutely continuous and Property N holds (see Section 3.1.4 and Theorem 3.12, which means that the set of rays r where μ^r has an atom is negligible). Hence $\nu^r_{|S}$ is a single Dirac mass. The mass equilibrium condition between μ^r and ν^r implies that the value of this mass must be equal to the difference $1-\nu^r_{|S^c}(r)$, and this last quantity does not depend on γ but only on μ and ν.

Finally, this proves that each $\mathbf{w}_{\gamma^r}$ does not depend on the choice of γ. □ □

Corollary 4.15. *If $\mu\ll\mathscr{L}^d$, then the solution of* (BP) *is unique.*

Proof. We have seen in Theorem 4.13 that any optimal $\mathbf{w}$ is of the form $\mathbf{w}_{[\gamma]}$ and in Theorem 4.14 that all the fields $\mathbf{w}_{[\gamma]}$ coincide. □ □

4.3 Summability of the transport density

The analysis of the Beckmann Problem performed in the previous sections was mainly made in a measure setting, and the optimal $\mathbf{w}$, as well as the transport density σ, were just measures on Ω. We investigate here the question whether they have extra regularity properties supposing extra assumptions on μ and/or ν.

We will give summability results, proving that σ is in some cases absolutely continuous and providing L^p estimates. The proofs are essentially taken from [273]: previous results, through very different techniques, were first presented in [144, 146, 145]. In these papers, different estimates on the "dimension" of σ were also presented, thus giving interesting information should σ fail to be absolutely continuous. Let us also observe that [209], while applying these tools to image processing problems, contains a new[4] proof of absolute continuity.

Note that higher order questions (such as whether σ is continuous or Lipschitz or more regular provided μ and ν have smooth densities) are completely open up to now. The only exception is a partial result in dimension 2, see [172], where a continuity result is given if μ and ν have Lipschitz densities on disjoint convex domains.

Open Problem (continuity of the transport density): is it true that, supposing that μ,ν have continuous densities, the transport density σ is also continuous? is it true that it is Lipschitz if they are Lipschitz, and/or Hölder if they are Hölder?

In all that follows Ω is a compact and convex domain in $\mathbb{R}^d$, and two probability measures μ,ν are given on it. At least one of them will be absolutely continuous, which implies uniqueness for σ (see Theorem 4.14).

Theorem 4.16. *Suppose $\Omega \subset \mathbb{R}^d$ is a convex domain, and $\mu \ll \mathscr{L}^d$. Let σ be the transport density associated to the transport of μ onto ν. Then $\sigma \ll \mathscr{L}^d$.*

Proof. Let γ be an optimal transport plan from μ to ν and take $\sigma = \sigma_\gamma$; call μ_t the standard interpolation between the two measures: $\mu_t = (\pi_t)_\#\gamma$ where $\pi_t(x,y) = (1-t)x+ty$: we have $\mu_0 = \mu$ and $\mu_1 = \nu$.

We have already seen (go back to (4.6)) that the transport density σ may be written as

$$\sigma = \int_0^1 (\pi_t)_\#(c\cdot\gamma)\mathrm{d}t,$$

where $c : \Omega\times\Omega \to \mathbb{R}$ is the cost function $c(x,y) = |x-y|$ (hence $c\cdot\gamma$ is a positive measure on $\Omega\times\Omega$).

Since Ω is bounded it is evident that we have

$$\sigma \le C\int_0^1 \mu_t\,\mathrm{d}t. \tag{4.11}$$

To prove that σ is absolutely continuous, it is sufficient to prove that almost every measure μ_t is absolutely continuous, so that, whenever $|A| = 0$, we have $\sigma(A) \le C\int_0^1 \mu_t(A)\mathrm{d}t = 0$.

[4] Their proof is somehow intermediate between that of [144] and the one we present here: indeed, approximation by atomic measures is also performed in [209], as here, but on both the source and the target measure, which requires a geometric analysis of the transport rays as in [144].